Task 1: Foundation & Environment Setup

Student: Lishajasmine Thangavel ID: APSPL2649129

1. Aim

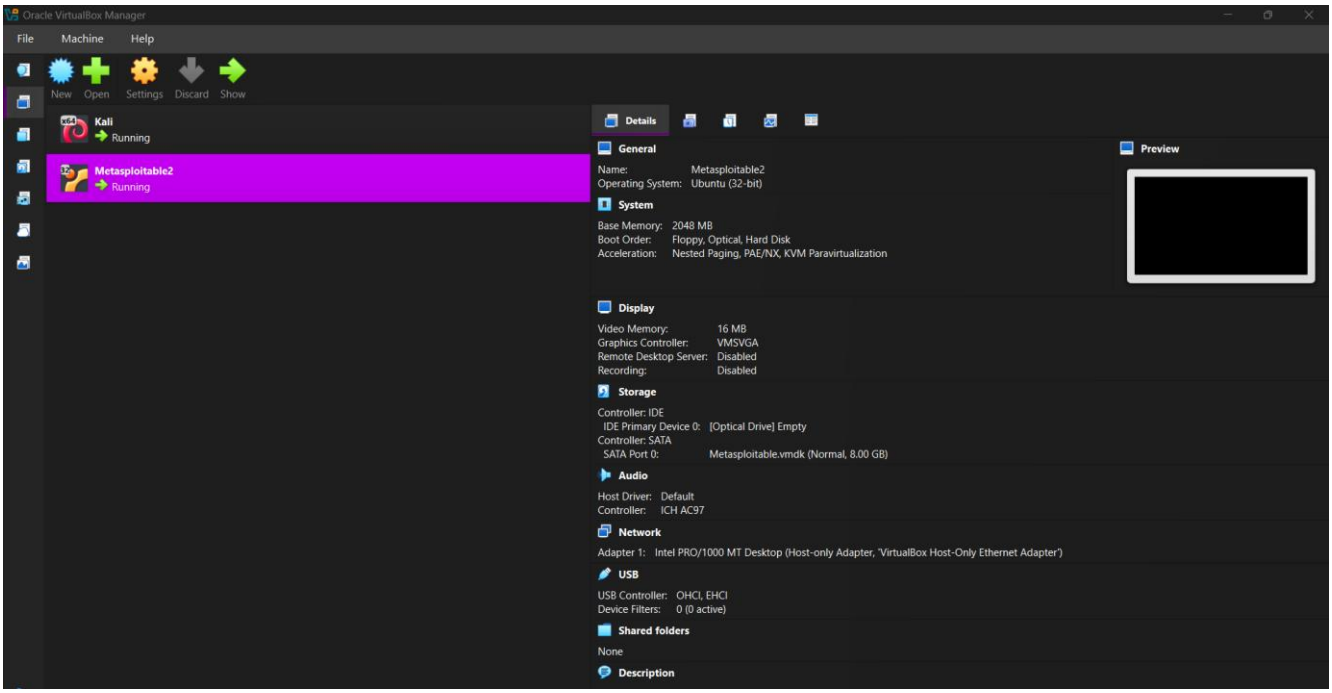
To prepare a controlled cybersecurity practice environment and demonstrate basic Linux file operations, network inspection, file encryption/decryption, packet capture, and verification of commonly used cybersecurity tools.

2. Lab Environment

Component	Observed configuration
Host system	Windows computer running Oracle VirtualBox
Attacker / analysis VM	Kali Linux
Target VM	Metasploitable2
Virtual network	VirtualBox Host-Only network
Host-Only adapter address	192.168.56.1/24
Kali Host-Only address (observed)	192.168.56.102/24 on eth1
Metasploitable2 address (observed)	192.168.56.101/24 on eth0
Packet analysis tool	Wireshark
Cryptography tool	OpenSSL
Other tools checked	Nmap, Burp Suite, Netcat

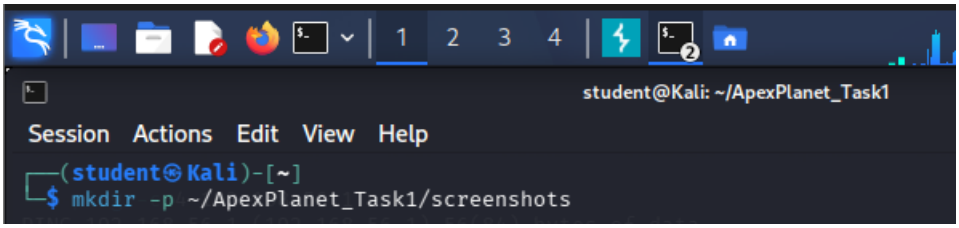
3. Practical Activities and Observations

3.1 Kali Linux and VirtualBox Host-Only Network Configuration



Observation: Oracle VirtualBox was used to manage the Kali Linux virtual machine and configure its network adapter. The Host-Only network provides a private communication channel for the cybersecurity lab.

3.2 Linux Directory and File Operations




```
(student㉿Kali)-[~]
$ cd ~/ApexPlanet_Task1
```

```
(student㉿Kali)-[~/ApexPlanet_Task1]
$ pwd
/home/student/ApexPlanet_Task1
```

```
(student㉿Kali)-[~/ApexPlanet_Task1]
$ echo "ApexPlanet Task 1 - Foundations and Environment Setup" > notes.txt
(student㉿Kali)-[~/ApexPlanet_Task1]
$ cat notes.txt
ApexPlanet Task 1 - Foundations and Environment Setup
```

```
(student㉿Kali)-[~/ApexPlanet_Task1]
$ ls -la
total 16
drwxrwxr-x  3 student student 4096 Oct 17 09:07 .
drwx------ 18 student student 4096 Oct 17 09:05 ..
-rw-rw-r--  1 student student   54 Oct 17 09:07 notes.txt
drwxrwxr-x  2 student student 4096 Oct 17 09:05 screenshots

(student㉿Kali)-[~/ApexPlanet_Task1]
$ echo "This is sample evidence for Task 1." > evidence.txt
```

```
(student㉿Kali)-[~/ApexPlanet_Task1]
$ cat evidence.txt
This is sample evidence for Task 1.

(student㉿Kali)-[~/ApexPlanet_Task1]
$ cp evidence.txt evidence_copy.txt

(student㉿Kali)-[~/ApexPlanet_Task1]
$ mv evidence_copy.txt evidence_renames.txt
```

```
(student㉿Kali)-[~/ApexPlanet_Task1]
$ ls -l
total 16
-rw-rw-r-- 1 student student   36 Oct 17 09:12 evidence_renames.txt
-rw-rw-r-- 1 student student   36 Oct 17 09:12 evidence.txt
-rw-rw-r-- 1 student student   54 Oct 17 09:07 notes.txt
drwxrwxr-x 2 student student 4096 Oct 17 09:05 screenshots

(student㉿Kali)-[~/ApexPlanet_Task1]
$ ls -l evidence.txt
-rw-rw-r-- 1 student student 36 Oct 17 09:12 evidence.txt
```

Observation: Linux commands were used to create directories and text files, display file contents, copy and rename files, and list directory contents. These operations demonstrated basic Linux file management.

3.3 Linux File Permissions Using chmod and ls -l

```
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ chmod 600 evidence.txt

┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ ls -l evidence.txt
-rw------- 1 student student 36 Oct 17 09:12 evidence.txt
```

Observation: The chmod command was used to change the permissions of a file, and ls -l was used to verify the resulting permissions. This demonstrated basic access control for files in Linux.

3.4 Network Interface and Routing Table Inspection in Kali Linux

```
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP
eth1             UP             192.168.56.102/24 fe80::a00:27ff:fef2:7fa3/64
```

```
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ ip route
192.168.56.0/24 dev eth1 proto kernel scope link src 192.168.56.102 metric 100

┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ ip addr show eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f2:7f:a3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
       valid_lft 399sec preferred_lft 399sec
    inet6 fe80::a00:27ff:fef2:7fa3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Observation: The ip -br addr and ip route commands were used to inspect network interfaces and routing information. The Kali Host-Only interface, eth1, displayed the IP address 192.168.56.102/24.

3.5 OpenSSL Encryption and Decryption Verification

```
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.021 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.048 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.021/0.042/0.051/0.012 ms

┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ ping -c 4 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.

--- 192.168.56.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3069ms
```



Observation: OpenSSL was used to encrypt a sample text file and decrypt the resulting encrypted file. The original and decrypted files were compared to verify that the file contents matched.

3.6 Wireshark Interface Selection and Packet Capture

```
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ mkdir -p ~/ApexPlanet_Task1/screenshots

┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ cd ~/ApexPlanet_Task1

┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ wireshark --version
Wireshark 4.6.6.

Copyright 1998-2026 Gerald Combs <gerald@wireshark.org> and contributors.
Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
 Bit width: 64-bit
  Compiler: GCC 15.2.0
      GLib: 2.88.1
 With:
  +brotli                      +nghttp2 1.69.0
  +Gcrypt 1.12.2               +nghttp3 1.15.0
  +GnuTLS 3.8.13 and PKCS#11   +PCRE2 10.46 2025-08-27
  +Kerberos (MIT)              +POSIX capabilities (Linux)
  +libnl 3                     +Qt 6.10.2
  +libpcap                     +QtDBus
  +libsmi 0.4.8                +QtMultimedia
  +libxml2 2.15.2              +Snappy 1.2.2
  +Lua 5.4.8                   +xxhash 0.8.3
  +LZ4 1.10.0                  +zlib 1.3.2
  +MaxMind                     +Zstandard 1.5.7
  +Minizip 1.3.2
 Without:
  -automatic updates  -zlib-ng

Runtime info:
      OS: Linux 6.19.14+kali-amd64
     CPU: 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz (with SSE4.2)
  Memory: 3917 MB of physical memory
    GLib: 2.88.1
  Locale: LC_TYPE=en_IN
 Plugins: supported, 0 loaded
 With:
  +brotli 1.2.0
  +c-ares 1.34.6
  +Gcrypt 1.12.2
  +GnuTLS 3.8.13
  +libpcap 1.10.6 (64-bit time_t, with TPACKET_V3)
```

```
 +brotli                  +nghttp2 1.69.0
 +Gcrypt 1.12.2           +nghttp3 1.15.0
 +GnuTLS 3.8.13 and PKCS#11  +PCRE2 10.46 2025-08-27
 +Kerberos (MIT)          +POSIX capabilities (Linux)
 +libnl 3                 +Qt 6.10.2
 +libpcap                 +QtDBus
 +libsmi 0.4.8            +QtMultimedia
 +libxml2 2.15.2          +Snappy 1.2.2
 +Lua 5.4.8               +xxhash 0.8.3
 +LZ4 1.10.0              +zlib 1.3.2
 +MaxMind                 +Zstandard 1.5.7
 +Minizip 1.3.2
Without:
 -automatic updates  -zlib-ng

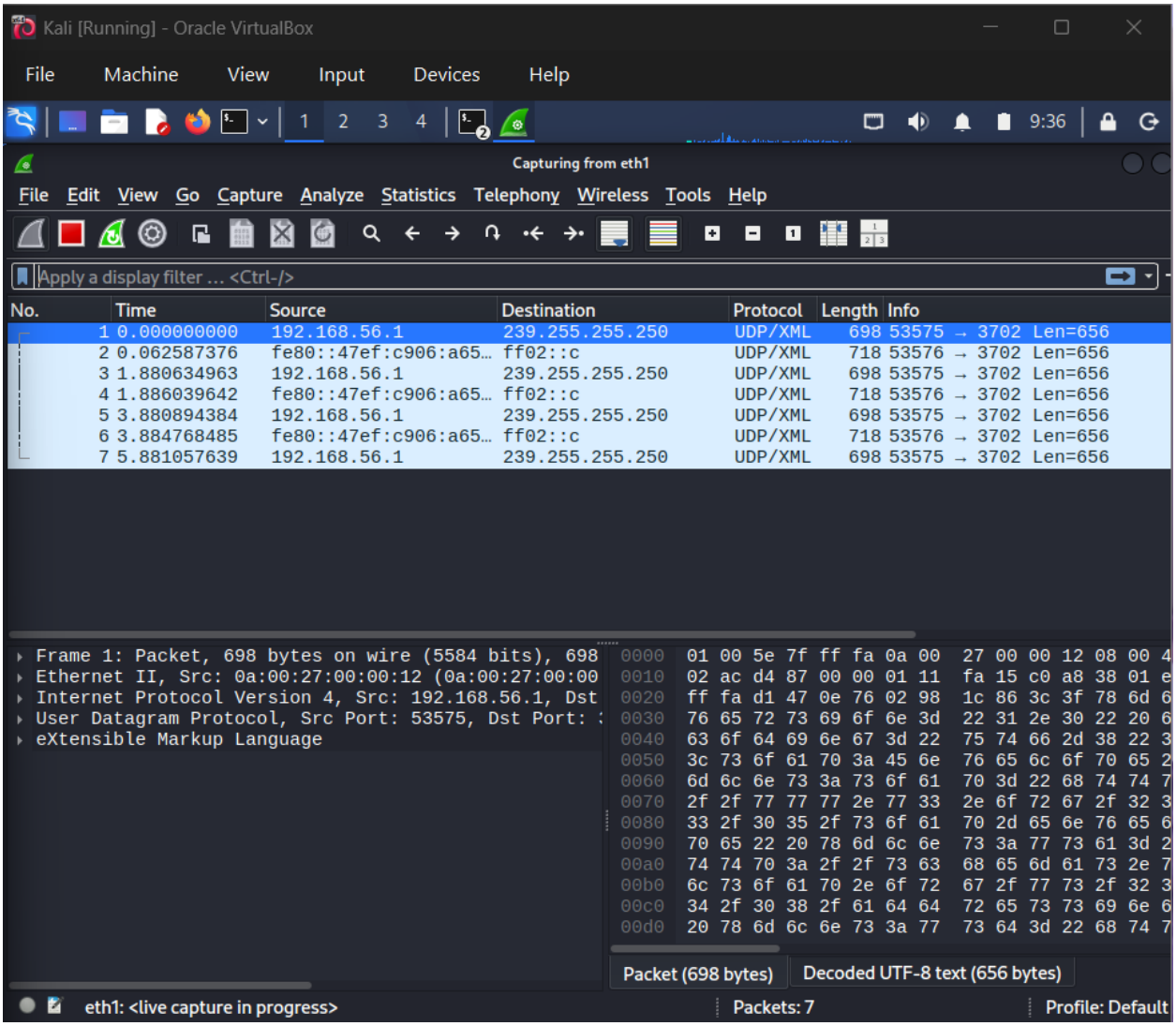
Runtime info:
     OS: Linux 6.19.14+kali-amd64
    CPU: 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz (with SSE4.2)
 Memory: 3917 MB of physical memory
   GLib: 2.88.1
 Locale: LC_TYPE=en_IN
Plugins: supported, 0 loaded
With:
 +brotli 1.2.0
 +c-ares 1.34.6
 +Gcrypt 1.12.2
 +GnuTLS 3.8.13
 +libpcap 1.10.6 (64-bit time_t, with TPACKET_V3)
 +libsmi 0.4.8
 +LZ4 1.10.0
 +nghttp2 1.69.0
 +nghttp3 1.15.0
 +PCRE2 10.46 2025-08-27
 +Qt 6.10.2
 +xxhash 803
 +zlib 1.3.2
 +Zstandard 1.5.7

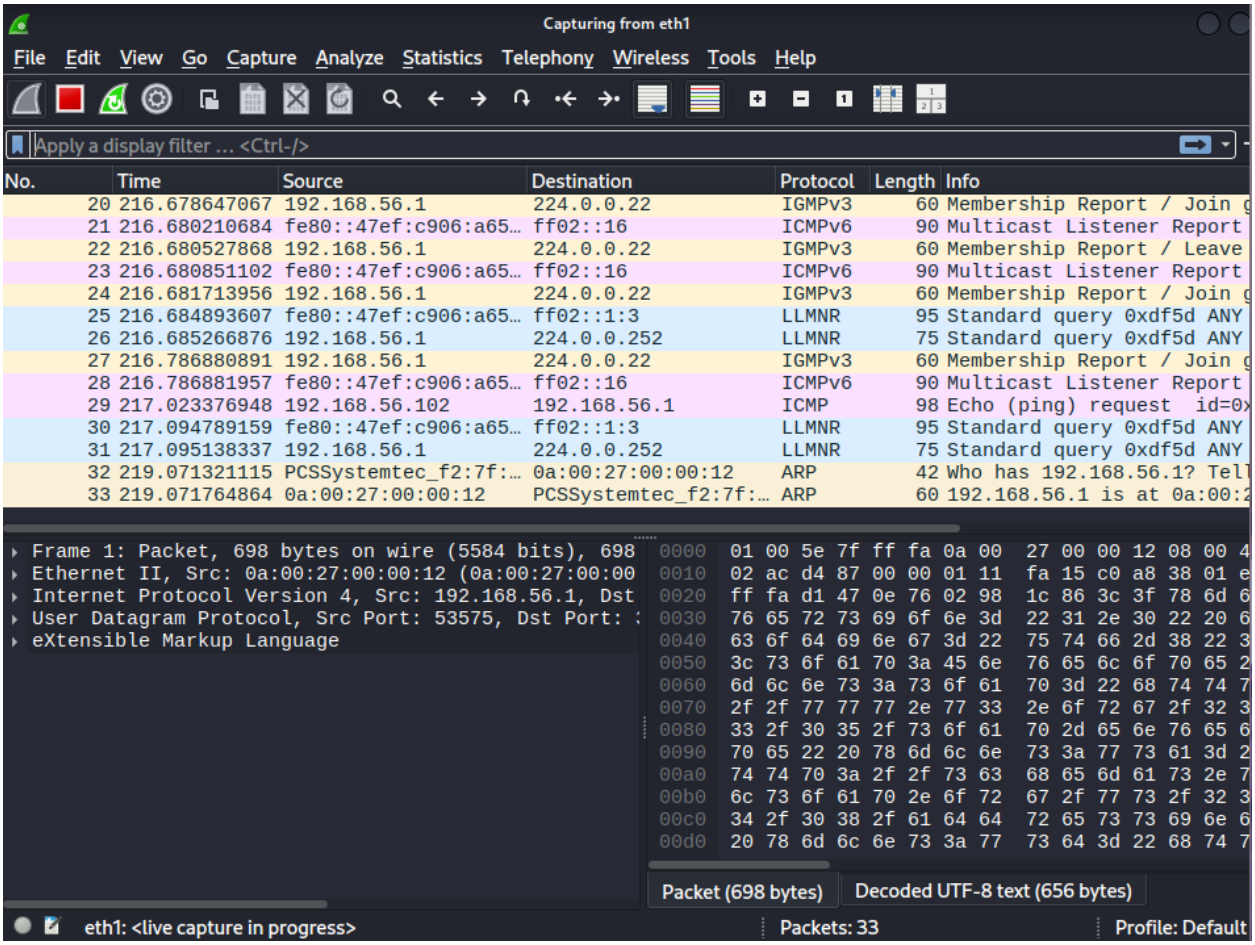
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ wireshark
** (wireshark:18468) 09:36:06.939551 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesk
top.portal.Error.Failed", "Could not register app ID: Connection already associated with an application ID")
** (wireshark:18468) 09:36:16.278909 [Capture MESSAGE] -- Capture Start ...
** (wireshark:18468) 09:36:16.311543 [Capture MESSAGE] -- Capture started
** (wireshark:18468) 09:36:16.311581 [Capture MESSAGE] -- File: "/tmp/wireshark_eth18B9FX3.pcapng"
** (wireshark:18468) 09:41:51.797040 [GUI WARNING] -- failed to create compose table
** (wireshark:18468) 09:45:12.994659 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:18468) 09:45:13.024926 [Capture MESSAGE] -- Capture stopped.
```

```
student@Kali: ~
Session  Actions  Edit  View  Help
┌──(student㉿Kali)-[~]
└─$ ping -c 4 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.

── 192.168.56.1 ping statistics ──
4 packets transmitted, 0 received, 100% packet loss, time 3056ms

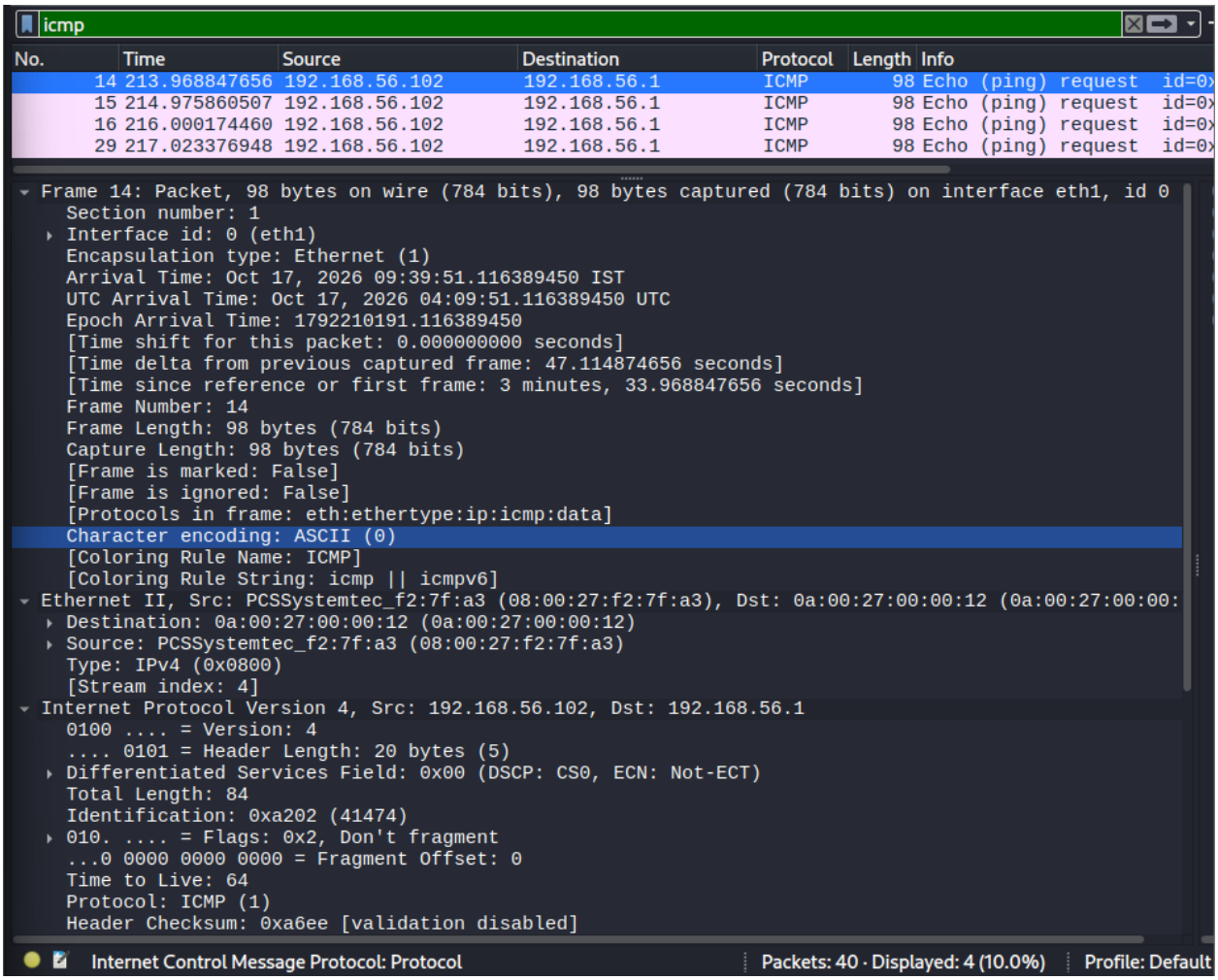
┌──(student㉿Kali)-[~]
```

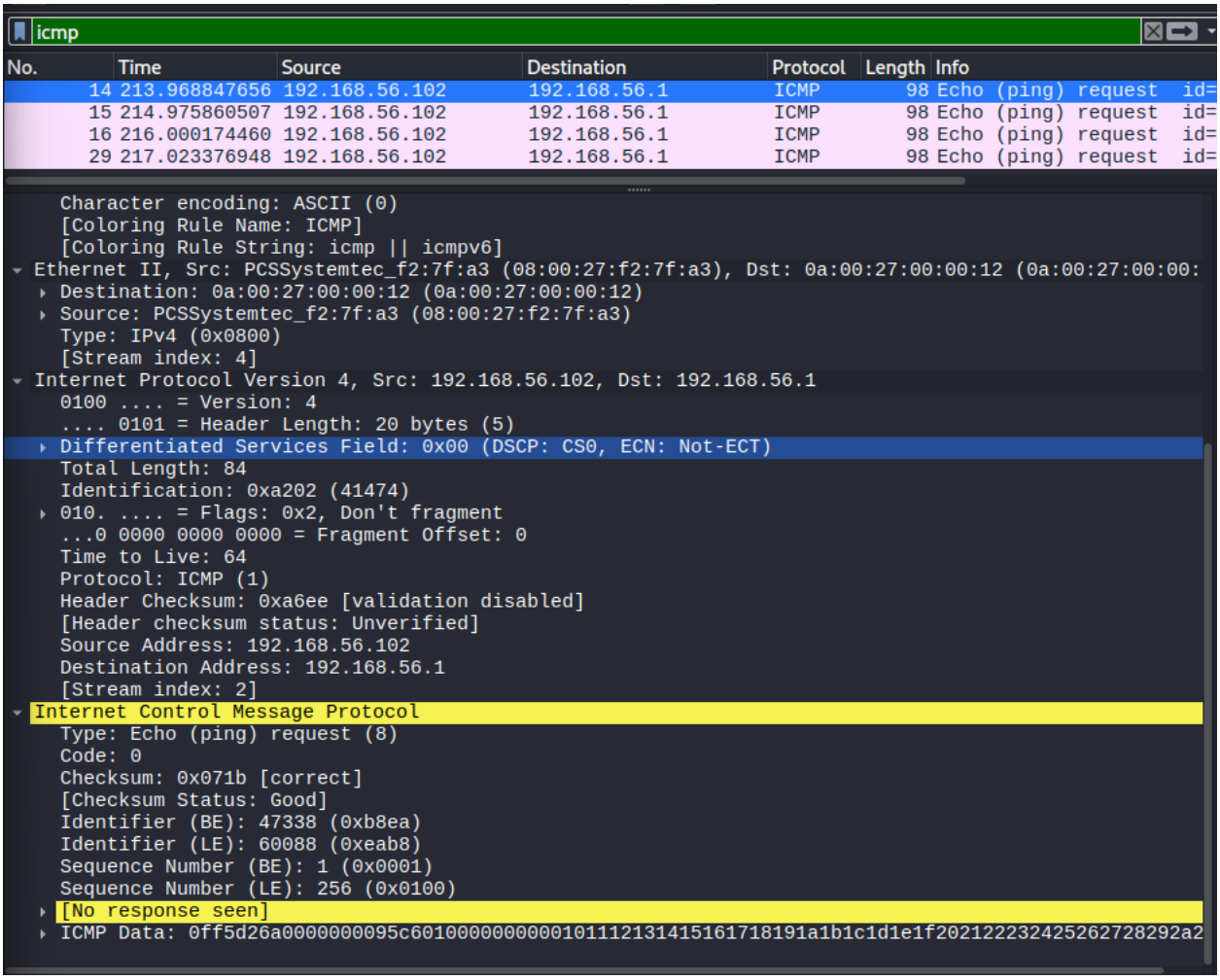

Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Capturing from eth1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 192.168.56.1 239.255.255.250 UDP/XML 698 53575 → 3702 Len=656
2 0.062587376 fe80::47ef:c906:a65… ff02::c UDP/XML 718 53576 → 3702 Len=656
3 1.880634963 192.168.56.1 239.255.255.250 UDP/XML 698 53575 → 3702 Len=656
4 1.886039642 fe80::47ef:c906:a65… ff02::c UDP/XML 718 53576 → 3702 Len=656
5 3.880894384 192.168.56.1 239.255.255.250 UDP/XML 698 53575 → 3702 Len=656
6 3.884768485 fe80::47ef:c906:a65… ff02::c UDP/XML 718 53576 → 3702 Len=656
7 5.881057639 192.168.56.1 239.255.255.250 UDP/XML 698 53575 → 3702 Len=656
Frame 1: Packet, 698 bytes on wire (5584 bits), 698
Ethernet II, Src: 0a:00:27:00:00:12 (0a:00:27:00:00
Internet Protocol Version 4, Src: 192.168.56.1, Dst
User Datagram Protocol, Src Port: 53575, Dst Port:
eXtensible Markup Language
0000 01 00 5e 7f ff fa 0a 00 27 00 00 12 08 00 4
0010 02 ac d4 87 00 00 01 11 fa 15 c0 a8 38 01 e
0020 ff fa d1 47 0e 76 02 98 1c 86 3c 3f 78 6d 6
0030 76 65 72 73 69 6f 6e 3d 22 31 2e 30 22 20 6
0040 63 6f 64 69 6e 67 3d 22 75 74 66 2d 38 22 3
0050 3c 73 6f 61 70 3a 45 6e 76 65 6c 6f 70 65 2
0060 6d 6c 6e 73 3a 73 6f 61 70 3d 22 68 74 74 7
0070 2f 2f 77 77 77 2e 77 33 2e 6f 72 67 2f 32 3
0080 33 2f 30 35 2f 73 6f 61 70 2d 65 6e 76 65 6
0090 70 65 22 20 78 6d 6c 6e 73 3a 77 73 61 3d 2
00a0 74 74 70 3a 2f 2f 73 63 68 65 6d 61 73 2e 7
00b0 6c 73 6f 61 70 2e 6f 72 67 2f 77 73 2f 32 3
00c0 34 2f 30 38 2f 61 64 64 72 65 73 73 69 6e 6
00d0 20 78 6d 6c 6e 73 3a 77 73 64 3d 22 68 74 7
Packet (698 bytes)
Decoded UTF-8 text (656 bytes)
eth1: <live capture in progress>
Packets: 7
Profile: Default



Observation: Wireshark was used to select a network interface and capture network traffic generated during the lab exercise. This demonstrated the initial steps involved in network packet analysis.

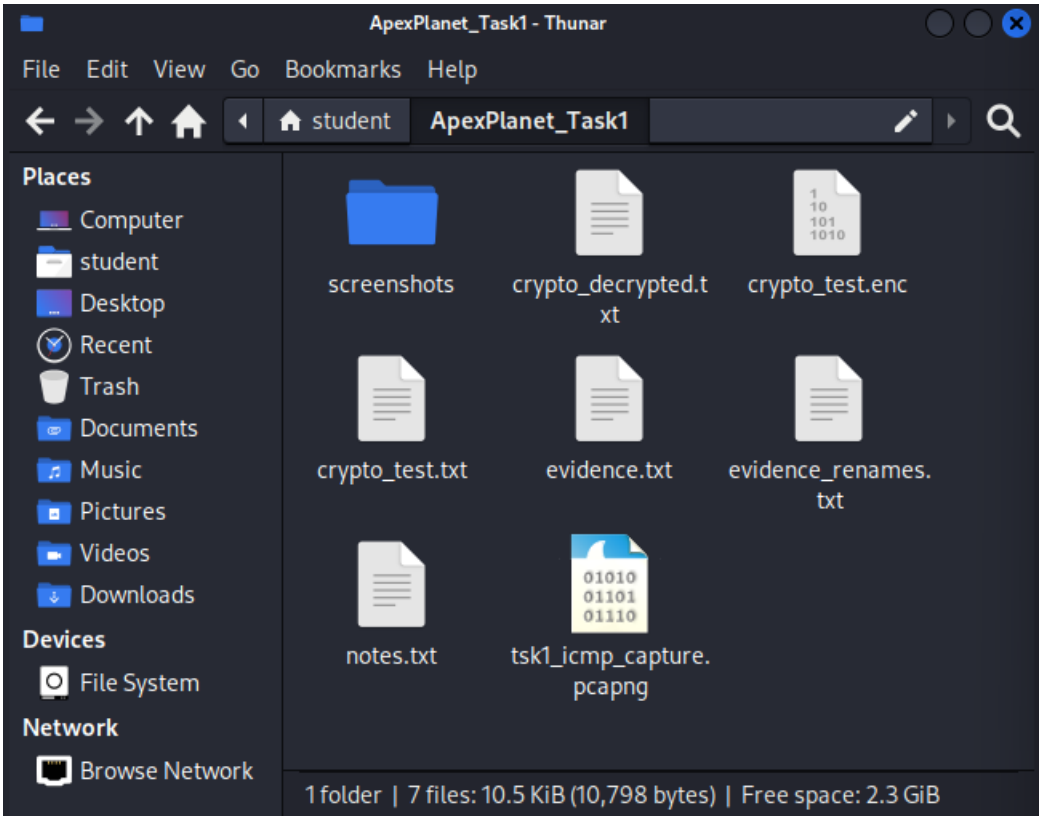
3.7 Wireshark ICMP Filtering and Packet Inspection





Observation: The icmp display filter was used to isolate ICMP packets generated during connectivity testing. Packet details were inspected to understand the information recorded in network traffic.

3.8 Saving the Wireshark Packet Capture File

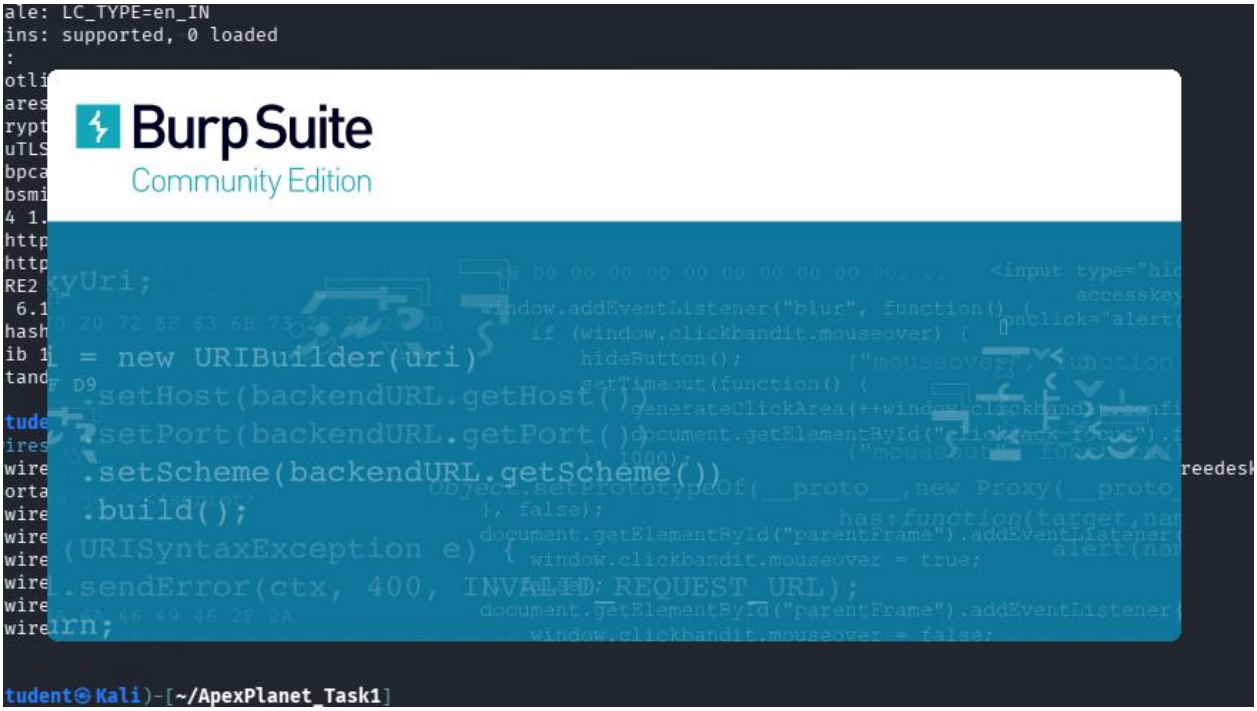


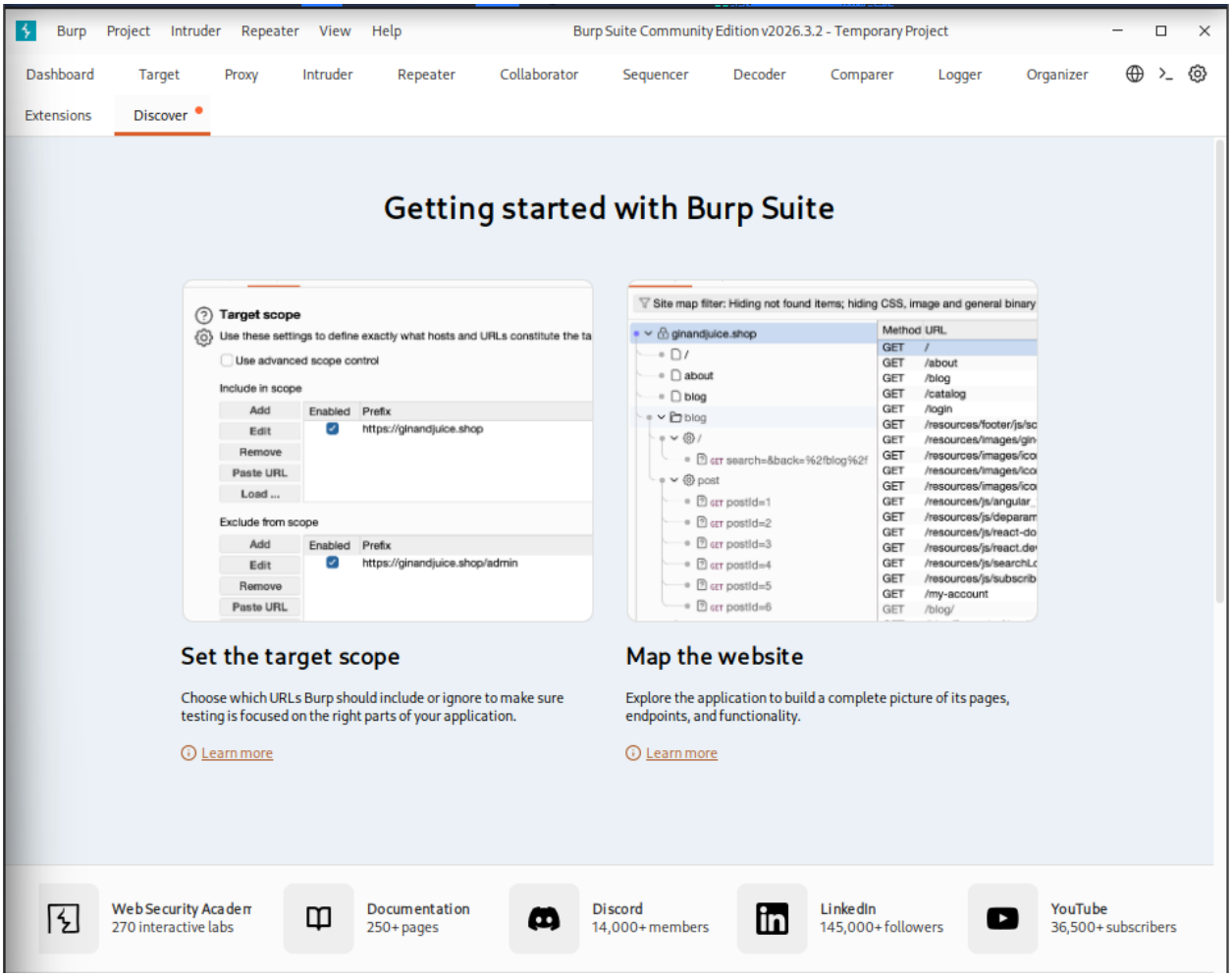
Observation: The Wireshark capture was saved in .pcapng format for later examination. The saved capture provides evidence of the network traffic observed during the practical exercise.

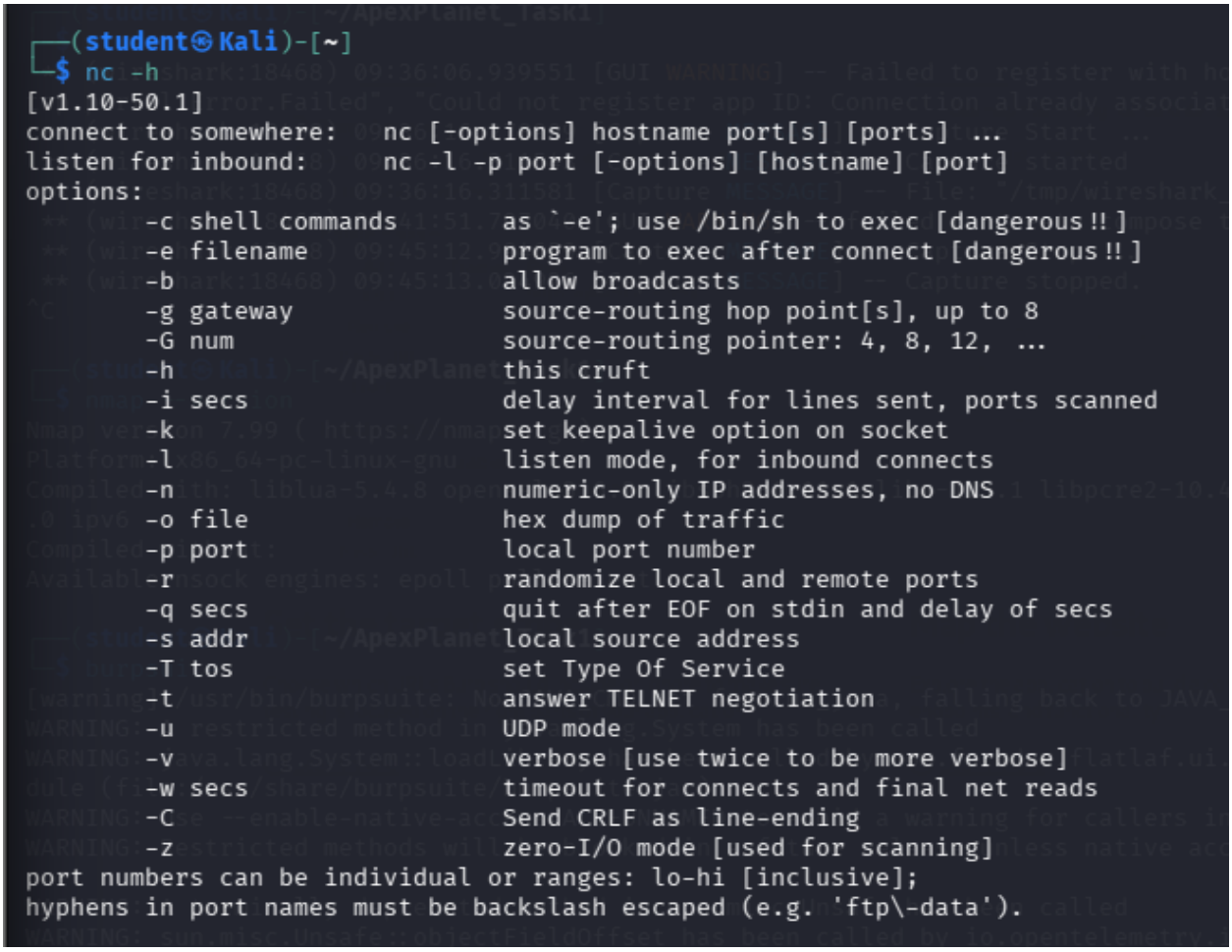
3.9 Verification of Cybersecurity Tools in Kali Linux

```
┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18
.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

┌──(student㉿Kali)-[~/ApexPlanet_Task1]
└─$ burpsuite
[warning] /usr/bin/burpsuite: No JAVA_CMD set for run_java, falling back to JAVA_CMD = java
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by com.formdev.flatlaf.ui.FlatNativeLibrary in an unnamed mo
dule (file:/usr/share/burpsuite/burpsuite.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
```


Burp Suite
Community Edition
tudent㉿Kali)-[~/ApexPlanet_Task1]


Burp Project Intruder Repeater View Help
Burp Suite Community Edition v2026.3.2 - Temporary Project
Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer
Extensions Discover
Getting started with Burp Suite
Target scope
Use these settings to define exactly what hosts and URLs constitute the ta
Use advanced scope control
Include in scope
Add
Edit
Remove
Paste URL
Load ...
Enabled
Prefix
https://ginandjuice.shop
Exclude from scope
Add
Edit
Remove
Paste URL
Enabled
Prefix
https://ginandjuice.shop/admin
Site map filter: Hiding not found items; hiding CSS, image and general binary
ginandjuice.shop
Method URL
GET /
GET /about
GET /blog
GET /catalog
GET /login
GET /my-account
Set the target scope
Choose which URLs Burp should include or ignore to make sure testing is focused on the right parts of your application.
Learn more
Map the website
Explore the application to build a complete picture of its pages, endpoints, and functionality.
Learn more
Web Security Academy
270 interactive labs
Documentation
250+ pages
Discord
14,000+ members
LinkedIn
145,000+ followers
YouTube
36,500+ subscribers



Observation: Nmap, Burp Suite, Netcat and OpenSSL were checked to confirm their availability. Burp Suite was launched using a temporary in-memory project, while the terminal commands were used to check the other tools.

3.10 Metasploitable2 Network Interface and IP Address Configuration

```
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ /sbin/ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:6b:ce:62
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe6b:ce62/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:5 errors:0 dropped:0 overruns:0 frame:0
          TX packets:53 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1910 (1.8 KB)  TX bytes:7807 (7.6 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:122 errors:0 dropped:0 overruns:0 frame:0
          TX packets:122 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:33885 (33.0 KB)  TX bytes:33885 (33.0 KB)

msfadmin@metasploitable:~$
```

Observation: The Metasploitable2 network interface was inspected using /sbin/ifconfig. The eth0 interface displayed the IP address 192.168.56.101 with the subnet mask 255.255.255.0, confirming its Host-Only network configuration.

3.11 Connectivity Verification Between Kali Linux and Metasploitable2

```
┌──(student㉿Kali)-[~]
└─$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP
eth1             UP             192.168.56.102/24 fe80::a00:27ff:fef2:7fa3/64
```

```
┌──(student㉿Kali)-[~]
└─$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=2.91 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.813 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.01 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.699 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3026ms
rtt min/avg/max/mdev = 0.699/1.609/2.914/0.912 ms
```

Observation: The ping command was used from Kali Linux to test connectivity with the Metasploitable2 target at 192.168.56.101. Replies were received, confirming communication between the two virtual machines over the Host-Only network.

4. Conclusion

The foundational lab activities were carried out in Kali Linux. Linux file operations and permission management were practised; network interfaces, routes and sockets were inspected; a sample file was encrypted and decrypted with OpenSSL; Wireshark was used to capture and inspect test traffic; and the availability of key cybersecurity tools was checked. Metasploitable2 was configured as a target VM on the Host-Only network, and the user confirmed successful connectivity from Kali to the target. The report should be finalised by inserting the original screenshots and checking that the saved capture file and repository contents are present.